

L^AT_EX Cheat Sheet

Document classes

`article` Best suited for exercise sheets.

Used at the very beginning of a document : `\documentclass{class}`.
Use `\begin{document}` to start contents and `\end{document}` to end the document.

Common documentclass options

10pt/11pt/12pt	Font size.
letterpaper/a4paper	Paper size.
twocolumn	Use two columns.
twoside	Set margins for two-sided.
landscape	Landscape orientation.
draft	Double-space lines.

Usage : `\documentclass[opt, opt]{class}`.

Recommended packages

N.B. In the PDF version of this cheat sheet all package names link to their respective CTAN page for quick access to the full documentation.

<code>babel</code>	Support for non-English languages. See below for French typesetting.
<code>geometry</code>	Page layout (margins, etc.)
<code>multicol</code>	Use <i>n</i> columns : <code>\begin{multicols}{n}</code> .
<code>fancyhdr</code>	Customize header and footer
<code>mathtools</code>	Advanced math typesetting. Includes <code>amsmath</code> .
<code>graphicx</code>	Show image : <code>\includegraphics[width=x]{file}</code> .
<code>hyperref</code>	Insert URL : <code>\url{https://...}</code> or hyperlink : <code>\href{URL}{text}</code> .
<code>enumitem</code>	Customize <code>enumerate</code> .
<code>tabularx</code>	Stretchable X columns in tables.
<code>tabulary</code>	Automatically balanced column widths.
<code>tabto</code>	Tab stops (compatible with enumerations).
<code>tikz</code>	TikZ ist kein Zeichenprogramm. Make beautiful and precise scientific drawings. Full online documentation : https://tikz.dev

`tkz-euclide` Tools for drawing Euclidean geometry; based on TikZ/PGF.

`tkz-tab` Tableaux de variations / signes.

`pgfplots` Create various high-quality plots; based on TikZ/PGF.

`beamer` For creating slides.
Use before `\begin{document}`.
Usage : `\usepackage{package}`

Title

`\author{text}` Author of document.
`\title{text}` Title of document.
`\date{text}` Date.

These commands go before `\begin{document}`. The declaration `\maketitle` goes at the top of the document.

Miscellaneous

`\pagestyle{empty}` Empty header, footer and no page numbers.
`\tableofcontents` Add a table of contents here.

Document structure

<code>\part{title}</code>	<code>\subsubsection{title}</code>
<code>\chapter{title}</code>	<code>\paragraph{title}</code>
<code>\section{title}</code>	<code>\subparagraph{title}</code>
<code>\subsection{title}</code>	

Use a `*`, as in `\section*{title}`, to not number a particular item—these items will also not appear in the table of contents.

Text environments

<code>\begin{comment}</code>	Comment (not printed). Requires <code>verbatim</code> package.
<code>\begin{quote}</code>	Indented quotation block.
<code>\begin{quotation}</code>	Like quote with indented paragraphs.
<code>\begin{verse}</code>	Quotation block for verse.

Lists

<code>\begin{enumerate}</code>	Numbered list.
<code>\begin{itemize}</code>	Bulleted list.
<code>\begin{description}</code>	Description list.
<code>\item text</code>	Add an item.
<code>\item[x] text</code>	Use <i>x</i> instead of normal bullet or number. Required for descriptions.

References

<code>\label{marker}</code>	Set a marker for cross-reference, often of the form <code>\label{sec:item}</code> .
<code>\ref{marker}</code>	Give section/body number of marker.
<code>\pageref{marker}</code>	Give page number of marker.
<code>\footnote{text}</code>	Print footnote at bottom of page.

Floating bodies

<code>\begin{table}[place]</code>	Add numbered table.
<code>\begin{figure}[place]</code>	Add numbered figure.
<code>\begin{equation}[place]</code>	Add numbered equation.
<code>\caption{text}</code>	Caption for the body.

The *place* is a list valid placements for the body. t=top, h=here, b=bottom, p=separate page, !=place even if ugly. Captions and label markers should be within the environment.

Text properties

Font face

Command	Declaration	Effect
<code>\textrm{text}</code>	<code>{\rmfamily text}</code>	Roman family
<code>\textsf{text}</code>	<code>{\sffamily text}</code>	Sans serif family
<code>\texttt{text}</code>	<code>{\ttfamily text}</code>	Typewriter family
<code>\textmd{text}</code>	<code>{\mdseries text}</code>	Medium series
<code>\textbf{text}</code>	<code>{\bfseries text}</code>	Bold series
<code>\textup{text}</code>	<code>{\upshape text}</code>	Upright shape
<code>\textit{text}</code>	<code>{\itshape text}</code>	<i>Italic shape</i>
<code>\textsl{text}</code>	<code>{\slshape text}</code>	<i>Slanted shape</i>
<code>\textsc{text}</code>	<code>{\scshape text}</code>	SMALL CAPS SHAPE
<code>\emph{text}</code>	<code>{\em text}</code>	<i>Emphasized</i>
<code>\textnormal{text}</code>	<code>{\normalfont text}</code>	Document font
<code>\underline{text}</code>		<u>Underline</u>

The command (*ttt*) form handles spacing better than the declaration (*ttt*) form.

Font size

<code>\tiny</code>	tiny	<code>\Large</code>	Large
<code>\scriptsize</code>	scriptsize	<code>\LARGE</code>	LARGE
<code>\footnotesize</code>	footnotesize		
<code>\small</code>	small	<code>\huge</code>	huge
<code>\normalsize</code>	normalsize		
<code>\large</code>	large	<code>\Huge</code>	Huge

These are declarations and should be used in the form `{\small ...}`, or without braces to affect the entire document.

Verbatim text

<code>\begin{verbatim}</code>	Verbatim environment.
<code>\begin{verbatim*}</code>	Spaces are shown as <code>_</code> .
<code>\verb!text!</code>	Text between the delimiting characters (in this case '!') is verbatim.

Justification

Environment	Declaration
<code>\begin{center}</code>	<code>\centering</code>
<code>\begin{flushleft}</code>	<code>\raggedright</code>
<code>\begin{flushright}</code>	<code>\raggedleft</code>

Better typesetting (by enabling hyphenation) can be achieved using the package `ragged2e` and its environments `RaggedRight`, `RaggedLeft`, and `Center`.

Miscellaneous

`\linespread{x}` changes the line spacing by the multiplier *x*.

French typography

Use `babel-french` :

```
\usepackage[french]{babel}
% fix spacing when using « » in source
\frenchsetup{og=«, fg=»}
% fix spacing in e.g. $1,23$
\usepackage{icomma}
```

1 ^{er}	1\ier	1 ^{ers}	1\iers	1 ^{re}	1\iere	1 ^{res}	1\ieres
2 ^e	2\ieme	2 ^{es}	2\iemes	M ^{me}	M\up{me}	D ^r	D\up{r}
œ	\oe	Œ	\OE	æ	\ae	Æ	\AE
n ^o	\no	n ^{os}	\nos	N ^o	\No	N ^{os}	\Nos
«	\og{}	»	\fg{}	« Oh! »	\frquote{Oh!}		

Polyglossia is a modern replacement for `babel`, but it's only compatible with XeLaTeX and LuaLaTeX.

Text-mode symbols

Symbols

<code>&</code>	<code>\&</code>	<code>-</code>	<code>_</code>	<code>...</code>	<code>\ldots</code>	<code>•</code>	<code>\textbullet</code>
<code>\$</code>	<code>\\$</code>	<code>^</code>	<code>\~{}</code>	<code> </code>	<code>\textbar</code>	<code>\</code>	<code>\textbackslash</code>
<code>%</code>	<code>\%</code>	<code>~</code>	<code>\~{}</code>	<code>#</code>	<code>\#</code>	<code>\$</code>	<code>\\$</code>

Delimiters

<code>'</code>	<code>\`</code>	<code>“</code>	<code>\`</code>	<code>{</code>	<code>\{</code>	<code>[</code>	<code>\[</code>	<code>(</code>	<code>\(</code>	<code><</code>	<code>\textless</code>
<code>'</code>	<code>\`</code>	<code>”</code>	<code>\`</code>	<code>}</code>	<code>\}</code>	<code>]</code>	<code>\]</code>	<code>)</code>	<code>\)</code>	<code>></code>	<code>\textgreater</code>

Dashes

Name	Source	Example	Usage
hyphen	-	X-ray	In words.
en-dash	--	1–5	Between numbers.
em-dash	---	Yes—or no?	Punctuation.

Line and page breaks

<code>\</code>	Begin new line without new paragraph.
<code>*</code>	Prohibit pagebreak after linebreak.
<code>\pagebreak</code>	Start new page.
<code>\columnbreak</code>	Start new column.
<code>\noindent</code>	Do not indent current line.

Spaces and rules

<code>~</code>	Fixed space, disallow linebreak (N. ~Biever).
<code>\smallskip</code>	Small vertical space.
<code>\medskip</code>	Medium vertical space.
<code>\bigskip</code>	Big vertical space.
<code>\hspace{l}</code>	Horizontal space of length l (Ex: $l = 20\text{pt}$).
<code>\vspace{l}</code>	Vertical space of length l .
<code>\rule{w}{h}</code>	Line of width w and height h .
Rules of width/height 0 are sometimes useful for solving spacing or alignment issues.	

Tables and tab stops

Tab stops: tabbing vs. tabto

`\=` Set tab stop. `\>` Go to tab stop.
Tab stops can be set on “invisible” lines with `\kill` at the end of the line. Normally `\\` is used to separate lines.

The tabbing environment is incompatible with `enumerate/itemize`. In this case it’s better to use the package `tabto` instead:

```
\TabPositions{2cm, 5cm}
Some\tab sample\tab text.
```

tabular environment

```
\begin{array}[pos]{cols}
\begin{tabular}[pos]{cols}
\begin{tabular*}{width}[pos]{cols}
```

tabular column specification

`l` Left-justified column.
`c` Centered column.
`r` Right-justified column.
`p{width}` Same as `\parbox[t]{width}`.
`@{decl}` Insert *decl* instead of inter-column space.
`|` Inserts a vertical line between columns.

tabular elements

`\hline` Horizontal line between rows.
`\cline{x-y}` Horizontal line across columns x through y .
`\multicolumn{n}{cols}{text}`
A cell that spans n columns, with $cols$ column specification.

Math mode

For inline math, use `\(...\)` or `$....$`. For displayed math, use `\[...\]` or `\begin{equation}`.
 2^x `2^{x}` x_0 `x_{0}`
 $\frac{x}{y}$ `\frac{x}{y}` $\sqrt[n]{x}$ `\sqrt[n]{x}`
 $\sum_{k=1}^n$ `\sum_{k=1}^n` $\prod_{k=1}^n$ `\prod_{k=1}^n`
 $\int_a^b f$ `\int_a^b f` $\iint_S$ `\iint_S`
 $\vec{u}$ `\vec{u}` $\overrightarrow{AB}$ `\overrightarrow{AB}`

Vectors, matrices, and determinants

`\vec{u}`
`\binom{4}{2}` $\vec{u}$
`\vec{v}`
`\begin{psmallmatrix}`
`-3 \ 1 \ 4`
`\end{psmallmatrix}`
`\begin{pmatrix}`
`1&2 \ 3&4`
`\end{pmatrix}`
`\begin{vmatrix}`
`1&2 \ 3&4`
`\end{vmatrix}`
 $\vec{v} \begin{pmatrix} -3 \\ 1 \\ 4 \end{pmatrix}$
 $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$
 $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$

Math-mode symbols

$\leq$	<code>\leq</code>	$\geq$	<code>\geq</code>	$\neq$	<code>\neq</code>	$\approx$	<code>\approx</code>
$\times$	<code>\times</code>	$\div$	<code>\div</code>	$\pm$	<code>\pm</code>	$\cdot$	<code>\cdot</code>
$^\circ$	<code>\circ</code>	$\circ$	<code>\circ</code>	$'$	<code>\prime</code>	$\cdots$	<code>\cdots</code>
∞	<code>\infty</code>	$\neg$	<code>\neg</code>	$\wedge$	<code>\wedge</code>	$\vee$	<code>\vee</code>
$\supset$	<code>\supset</code>	$\forall$	<code>\forall</code>	$\in$	<code>\in</code>	$\rightarrow$	<code>\rightarrow</code>
$\subset$	<code>\subset</code>	$\exists$	<code>\exists</code>	$\notin$	<code>\notin</code>	$\Rightarrow$	<code>\Rightarrow</code>
$\cup$	<code>\cup</code>	$\cap$	<code>\cap</code>	$ $	<code>\mid</code>	$\Leftrightarrow$	<code>\Leftrightarrow</code>
$\dot{a}$	<code>\dot{a}</code>	$\hat{a}$	<code>\hat{a}</code>	$\bar{a}$	<code>\bar{a}</code>	$\tilde{a}$	<code>\tilde{a}</code>
α	<code>\alpha</code>	β	<code>\beta</code>	γ	<code>\gamma</code>	δ	<code>\delta</code>
ϵ	<code>\epsilon</code>	ζ	<code>\zeta</code>	η	<code>\eta</code>	ε	<code>\varepsilon</code>
θ	<code>\theta</code>	ι	<code>\iota</code>	κ	<code>\kappa</code>	ϑ	<code>\vartheta</code>
λ	<code>\lambda</code>	μ	<code>\mu</code>	ν	<code>\nu</code>	ξ	<code>\xi</code>
π	<code>\pi</code>	ρ	<code>\rho</code>	σ	<code>\sigma</code>	τ	<code>\tau</code>
υ	<code>\upsilon</code>	ϕ	<code>\phi</code>	χ	<code>\chi</code>	ψ	<code>\psi</code>
ω	<code>\omega</code>	Γ	<code>\Gamma</code>	Δ	<code>\Delta</code>	Θ	<code>\Theta</code>
Λ	<code>\Lambda</code>	Ξ	<code>\Xi</code>	Π	<code>\Pi</code>	Σ	<code>\Sigma</code>
Υ	<code>\Upsilon</code>	Φ	<code>\Phi</code>	Ψ	<code>\Psi</code>	Ω	<code>\Omega</code>

Sample L^AT_EX document

```
\documentclass[a4paper,11pt]{article}
% compile with LuaLaTeX
\usepackage{mathtools}% loads amsmath
\usepackage{unicode-math}% loads fontspec

\title{Sample document}
\author{A. Uthor}

\begin{document}
\maketitle

\section{Section title}
\subsection*{Unnumbered subsection}
text \textbf{bold text} text. Some math:  $\$2+2=5\$$ 
\subsection{Subsection title}
text \emph{emphasized text} text. Display math:
\[
\int_0^1 x^2 \, dx = \frac{1}{3}
\]

A centered table:
\begin{center}
\begin{tabular}{|l|c|r|}
\hline
first & row & abc \\
second & row & defghi \\
\hline
\end{tabular}
\end{center}
\end{document}
```

Online tools

Detexify: <https://detexify.kirelabs.org/>
Table generator: https://www.tablesgenerator.com/latex_tables
Mathpix: <https://mathpix.com/>